

following command:

```
$ ansible-playbook -i inventory/kvm/inventory playbooks/
configuration/consul.yml --tags consul -vv -K
```

The ‘-vv’ represents the verbosity in the Ansible output. You can use up to four ‘v’s for a more detailed output. The ‘-K’ option prompts for the sudo password for the *debian* user account.

You can also now log in to any one of the nodes and check the version output by running Consul as indicated below:

```
$ consul --version
Consul v1.1.0
```

```
Protocol 2 spoken by default, understands 2 to 3 (agent will
automatically use protocol >2 when speaking to compatible
agents)
```

Bootstrap node

The bootstrap Consul node is the first to be configured. The */etc/consul.d/bootstrap* directory is created and its configuration file (*bootstrap-config.json*) is copied to it. Its file contents are as follows:

```
{
  "bootstrap": true,
  "client_addr": "0.0.0.0",
  "server": true,
  "datacenter": "chennai",
  "data_dir": "/var/consul",
  "encrypt": "PfCkrq/lFPem7slKP5N2Cw==",
  "log_level": "INFO",
  "enable_syslog": true
}
```

The configuration specifies that this is a bootstrap node, and the server should bind to any IP address. We specify a name to the data centre, and also the path to the data directory. An encryption key is specified to encrypt the traffic between the Consul nodes. Finally, the log level and use of syslog is specified in the configuration file.

We also create a *systemd* configuration file for starting the bootstrap Consul node as shown below:

```
[Unit]
Description=Consul service discovery agent
Requires=network-online.target
After=network-online.target

[Service]
Environment=GOMAXPROCS=2
Restart=on-failure
ExecStart=/usr/local/bin/consul agent -server -ui -data-dir=
```

```
tmp/consul -node=host1 -config-dir=/etc/consul.d/bootstrap
-enable-script-checks=true
ExecReload=/bin/kill -HUP $MAINPID
KillSignal=SIGTERM
```

```
[Install]
WantedBy=multi-user.target
```

The ‘-ui’ option is provided to enable the Web front-end for Consul. The bootstrap node’s name is ‘host1’. The entire playbook to set up and start the bootstrap node is as follows:

```
- name: Configure bootstrap Consul node
  hosts: bootstrap
  become: yes
  become_method: sudo
  gather_facts: true
  tags: [bootstrap]

tasks:
  - name: Create Consul bootstrap directory
    file:
      path: /etc/consul.d/bootstrap
      state: directory
      mode: 0755

  - name: Copy configuration file
    copy:
      src: ../../files/bootstrap-config.json
      dest: /etc/consul.d/bootstrap/config.json

  - name: Copy systemd bootstrap consul.service
    copy:
      src: ../../files/bootstrap-consul.service
      dest: /etc/systemd/system/consul.service

  - name: Start bootstrap Consul
    systemd:
      name: consul.service
      state: started
```

A sample execution of the above playbook is given below:

```
$ ansible-playbook -i inventory/kvm/inventory playbooks/
configuration/consul.yml --tags bootstrap -K
SUDO password:

PLAY [Install Consul] *****

TASK [Gathering Facts] *****
*****ok: [host1]
ok: [host2]
ok: [host3]
```